

# TUSHAR VARSHNEY

github.com/tusharv2005 · tvarshney585@gmail.com · +91 70218 71011 · linkedin.com/tusharvarshney03 · Mumbai, India

## Summary

ML engineer and MBA Tech (AI) student at NMIMS Mumbai (GPA 3.6/4.0) building forecasting systems, production-grade tooling, and shipped products end-to-end. Strongest proof: an AI trading system delivering **18% alpha** vs. benchmark and a 40+ component open-source library at **85%+ test coverage**. Seeking internships in AI/ML engineering, applied research, or full-stack development.

## Technical Skills

|                 |                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| AI / ML         | PyTorch · HuggingFace Transformers · scikit-learn · CNN / ResNet · Transfer Learning · OpenCV · Time-Series Forecasting · Model Optimisation |
| Languages       | Python · TypeScript · JavaScript · C++ · SQL · C#                                                                                            |
| Frontend        | React · Tailwind CSS · Storybook · Jest / RTL · Unity                                                                                        |
| Backend & Data  | Node.js · REST APIs · PostgreSQL · Database Design · Server-side Logic                                                                       |
| Cloud & Tooling | AWS · Git · Tableau · Power BI · Excel (Advanced)                                                                                            |

## Projects

Eclipse Alpha Fund — AI Trading & Portfolio Optimisation System

Python · PyTorch · Time-Series

- Built a PyTorch time-series forecasting engine that outperformed benchmark indices by **18% simulated alpha**; validated robustness under distribution shift (regime changes, volatility spikes) via walk-forward cross-validation.
- Engineered an end-to-end ML pipeline — real-time API ingestion, feature engineering (20+ technical indicators, volatility metrics), multi-asset backtesting framework — cutting analysis runtime from **hours to under 3 minutes** through automated orchestration.
- Applied mean-variance and risk-adjusted optimisation to reduce maximum drawdown by **25%** while sustaining stable returns; designed failure-mode logging that surfaces violated model assumptions in real time — production-grade safety thinking applied to quant finance.

PetalUI — Open-Source React Component Library

React · TypeScript · Tailwind CSS · Storybook · Jest / RTL

- Architected and shipped a **40+ component** open-source UI library from scratch with zero external dependencies; WCAG 2.1 AA compliant across all interactive elements.
- Authored three custom React hooks (`useModal`, `useForm`, `useTheme`) that **cut per-feature boilerplate by 60%**; established Storybook + Jest/RTL test suite at **85%+ coverage**, eliminating a category of integration regressions.

Block Puzzle Master — Shipped Android Game

Unity · C# · Object Pooling · Finite State Machines

- Delivered the full game end-to-end as solo engineer; grid logic and FSM-based state machines maintained **stable 60 fps** across **50+ device profiles** with custom object-pooling cutting GC pauses by **40%** and peak memory by **35%**.
- Data-driven level pipeline enabled **10× faster content iteration**; A/B tested three difficulty curves, identifying the configuration that raised **Day-7 retention by 20%**.

## Experience

Founder & Technical Lead

Horizn Games — Independent Game Studio

2025 – Present

- Shipped 2 multi-platform titles end-to-end as sole technical lead: spec, prototype, QA, launch, and data-driven post-launch iteration.
- Scaled gaming community to **350,000+ followers** organically via algorithmic content strategy and SEO — **zero paid acquisition**.
- Cut bug-to-fix cycle from 5 days to **under 48 hours** by introducing lightweight Agile sprints and structured GitHub issue tracking.

AI Engineering Intern

Mahindra Rise (Mahindra & Mahindra)

May 2026 – Jul 2026

- Developed AI-powered enterprise search and conversational AI systems to enhance intelligent customer interaction and digital discovery experiences.
- Contributed to analytics-driven AI workflows and production-level product development focused on scalable enterprise AI solutions.
- Worked on intelligent automation features integrating semantic search, conversational interfaces, and AI-driven customer engagement technologies.

## Education

|                |                                                                          |               |
|----------------|--------------------------------------------------------------------------|---------------|
| 2024 – Present | MBA Tech (AI), NMIMS Mumbai — Artificial Intelligence & Entrepreneurship | GPA 3.6 / 4.0 |
| 2020           | X (SSC), Bharda New High School, MSBSHSE                                 | 90.8%         |